

TABLE II
Different E-Agriculture Projects in India

Project name	Website	Brief description
Application of electronics for agriculture and environment (eAgriEn)	http://eagrienv.cdackolkata.in/	Objectives of this ongoing flagship programme of the Department of Information Technology are: 1. To set up an integrated facility for advanced research in the field of agri-electronics leading to technology demonstrations and indigenous electronics and IT-based products and services for the Indian agricultural community 2. To provide IT-based services and consultancy to agricultural entrepreneurs, researchers and other stakeholders on specialised solutions developed under the aegis of the government 3. To conduct special training programmes on advanced agri-tech topics 4. To promote national- and international-level networking, collaboration and alliances in the field of agri-electronics
Web-enabled access of agricultural information (WEAAI) through PC and mobile devices	http://bau-eagriculture.com/submit/	The Web-based system is used to disseminate agricultural information to the farmers in the state of Jharkhand. Content is being developed in the areas of agriculture, animal husbandry and forestry in seven languages, namely, English, Hindi, Nagpuri, Santhali, Kurukh, Mundari and Ho. The solution involves vernacular Web-enabled interactive access, offline content delivery using LCMS, guided and unguided SMS in push as well as pull mode, and IVRS technologies for interactive and non-interactive dissemination of agricultural information.
Bhoomi projects	http://bhoomi.karnataka.gov.in/	Bhoomi is an integrated land record system in digitised form in Karnataka. Its main components are: 1. Mutation and updation of land records are done online at the village computer centre. It includes fingerprint authentication and scanning of important documents to ensure a robust and secured system. Most components of the computer centre are funded under the central scheme. 2. Land records kiosk where the farmers can collect the copy of their record by paying a minimal amount. They can also lodge request for mutation of their land records. The kiosk is fully-funded by the state government. 3. Touchscreen kiosk where farmers can see their land related information without anybody's intervention or help.
Gyandoot project	http://www.gyandoot.nic.in/	Gyandoot is an intranet in Dhar district (Madhya Pradesh) connecting rural cyber cafés catering to the everyday needs of the masses. The website of Gyandoot is an extension of the intranet for providing global access. It provides commodity/mandi marketing information system, income certificate, domicile certificate (<i>mool nivas</i>) and caste certificate, etc.
IFFCO-Airtel alliance		Indian Farmers' Fertilizer Cooperative (IFFCO) has partnered with Bharti Airtel to set up IFFCO Kisan Sanchar in Rajasthan. Airtel will provide communication services to farmers in the rural areas of the state with the help of Rajasthan Cooperative Department. Under this initiative, the department will provide mobile handsets to farmers at marginal price through its outlets in the rural areas. These handsets would be loaded with green SIM cards, which would flash daily updates. These flash messages would be related to commodity prices of the nearest mandi, weather forecast, usage of quality of seeds, fertilisers and information related to animal and plant diseases.
SEEDNET	http://seednet.gov.in/	This portal contains information on different types of seeds, seed companies, crop diseases, extension services, quality controls, etc. There are many organisations like NSRTC, SFCI, NSC, PPV&FRA, state governments, state seed corporations, certification agencies, testing laboratories, SHGs and cooperatives that work on different types of seeds and train the SHGs in different crop production in different seasons in different regions of India.
National e-Governance plan	http://india.gov.in/howdo/otherservice_details.php?service=6	To make farming communities benefit from the new global market access opportunities, the internal agricultural marketing system of the country is being integrated and strengthened. Through this portal, farmers can check agricultural market prices online.
mKrishi by TCS		With mKrishi, farmers can send their queries to a remote expert through their mobile CDMA handsets. Along with their queries, they can also send a photograph of the crop through a camera phone. The information related to crop, soil and micro-environment gathered by sensors is sent to experts through an automatic weather station using the cellular network. Farmers receive responses to their queries through the same channel. The end-to-end mKrishi solution has been developed through the integration of technologies such as sensors, solar power, CDMA modem and CDMA network, GPS, handset with camera, binary runtime environment for wireless to develop multiple applications on the handset, client software (on mobile phone), expert console software and an engine to assist in displaying mobile screens in Indian languages.
e-Choupal	http://www.echoupal.com/	ITC's agri-business division has conceived e-Choupal as a supply chain aimed at delivering value to its customers around the world on a sustainable basis. The e-Choupal model is designed to tackle the challenges posed by the features of Indian agriculture—fragmented farms, weak infrastructure and the involvement of numerous intermediaries, among others. With a blend of click and mortar capabilities, village Internet kiosks managed by farmers—called <i>sanchalaks</i> —enable the agricultural community to access information in their local language on weather and market prices. Also these help disseminate knowledge on scientific farm practices and risk management, and facilitate the sale of farm inputs (now with embedded knowledge) and purchase of farm produce from the farmers' doorsteps (since decision-making becomes information-based).
Reuters Market Light		The information marketing company has launched mobile-based personalised information provision services to Indian farmers. The service started in 2007 in Maharashtra, Punjab and Haryana. Presently, 12,000 villages are covered and more than 100,000 farmers are direct beneficiary of the service. 150 crops and 500 <i>mandals</i> are covered under the service. Farmers have to buy the service by paying a monthly subscription. They receive three to five SMS every day in vernacular language on the preferred topics and crops. Also, utility information like weather report, soil related information, world trends in agricultural production and prices are sent to the registered farmers.